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CHAPTER 16 — SHAREABILITY AND COHERENCY

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A maximum-detail, RTOS/QNX-style technical breakdown of ARMv8 shareability domains, cache coherency rules, multi-core interactions, memory ordering, and how Stage-1/Stage-2 descriptors, TCR, and MAIR combine to enforce correct SMP behavior.

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16.1 — Introduction to Shareability and Coherency

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Shareability and coherency determine how multiple cores:

- observe memory updates
- maintain cache consistency
- synchronize shared data
- interact with device memory
- perform atomic operations
- execute SMP-safe code

ARMv8 defines **shareability domains** that control how caches and memory systems behave across cores and clusters.

Shareability is encoded in:

- **SH** (Stage-1 descriptors)
- **S2SH** (Stage-2 descriptors)
- **TCR_ELx.SH0/SH1**
- **MAIR_ELx** (indirectly)

This chapter explains how these fields interact to produce correct SMP behavior.

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16.2 — What Is Shareability?

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Shareability defines whether memory accesses must be coherent across:

- cores
- clusters
- system interconnects

Shareability affects:

- cache maintenance
- TLB invalidation
- memory ordering
- atomic operations
- DMA behavior
- device access

Shareability is **not** the same as cacheability.

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16.3 — Shareability Domains

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ARMv8 defines three shareability domains:

Non-Shareable (SH = 00)

- memory is private to a core
- no coherency required
- no cache maintenance propagation
- used for device memory
- used for private scratch memory

Outer Shareable (SH = 10)

- coherent across clusters
- used for multi-cluster systems
- used for L3-coherent systems

Inner Shareable (SH = 11)

- coherent across cores in a cluster
- required for SMP correctness
- used for normal memory
- used for DMA buffers

Teaching Note

Most systems use **Inner Shareable** for all normal memory.

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16.4 — Shareability in Stage-1 Descriptors (SH)

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Stage-1 descriptors contain:

`SH[1:0]`

This defines shareability for:

- user memory
- kernel memory
- guest memory

Rules

- Normal memory → Inner Shareable
- Device memory → Non-Shareable
- DMA buffers → Inner Shareable
- Shared memory → Inner Shareable

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16.5 — Shareability in Stage-2 Descriptors (S2SH)

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Stage-2 descriptors contain:

`S2SH[1:0]`

Hypervisor uses S2SH to override guest shareability.

Example

Guest marks memory Non-Shareable:

```
SH = 00
```

Hypervisor overrides:

```
S2SH = 11 → Inner Shareable
```

Final shareability = Inner Shareable.

Teaching Note

Hypervisor always wins.

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16.6 — Shareability in TCR_ELx

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TCR_EL1 and TCR_EL2 contain:

- **SH0** → shareability for TTBR0
- **SH1** → shareability for TTBR1

These override descriptor shareability **only for page table walks**.

Important

TCR shareability does **not** override final memory shareability.

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16.7 — Shareability and Cache Coherency

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Shareability determines whether caches must:

- maintain coherence
- propagate invalidations
- propagate clean operations
- propagate TLB invalidations

Inner Shareable

- L1/L2 caches coherent
- TLB invalidations broadcast
- walk cache invalidations broadcast

Outer Shareable

- L3 coherent
- multi-cluster coherence

Non-Shareable

- no coherence
- no broadcast invalidations
- used for device memory

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16.8 — Shareability and Atomic Operations

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Atomic operations require:

- Normal memory
- Inner Shareable
- cacheable attributes

Illegal

Atomic operations on:

- Device memory
- Non-Shareable memory
- Non-cacheable memory

Teaching Note

Atomics on MMIO cause undefined behavior.

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16.9 — Shareability and Memory Ordering

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Shareability affects ordering:

Inner Shareable

- strong ordering within cluster
- required for SMP correctness

Outer Shareable

- ordering across clusters

Non-Shareable

- ordering only within core
- device memory ordering enforced separately

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16.10 — Shareability and DMA

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DMA engines operate outside CPU caches.

DMA Buffers

Must be:

- Normal memory
- Inner Shareable
- cacheable or non-cacheable depending on usage

MMIO Registers

Must be:

- Device-nGnRnE
- Non-Shareable

Cache Maintenance

Required for DMA:

- **DC CVAC**
- **DC CIVAC**
- **DSB SY**

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16.11 — Shareability and TLB Coherency

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TLB invalidations propagate based on shareability:

Inner Shareable

- broadcast to all cores in cluster
- required for SMP correctness

Outer Shareable

- broadcast across clusters

Non-Shareable

- no broadcast
- used for device memory

Implementation Note

TLBI must be followed by DSB SY + ISB.

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16.12 — Shareability and Walk Cache Coherency

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Walk cache stores intermediate descriptors.

Shareability determines:

- whether walk cache invalidations propagate
- whether stale descriptors remain
- whether BBM is safe

Rules

- Normal memory → Inner Shareable
- Device memory → Non-Shareable

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16.13 — Shareability and Speculation

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Speculation rules depend on shareability:

Inner Shareable

- speculation allowed
- prefetch allowed

Non-Shareable

- speculation forbidden
- prefetch forbidden

Teaching Note

Device memory must be Non-Shareable.

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16.14 — Shareability and Prefetch

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Prefetch behavior:

Normal Memory

- prefetch allowed
- improves performance

Device Memory

- prefetch forbidden
- may corrupt device state

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16.15 — Shareability and Stage-1/Stage-2 Merging

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Final shareability = Stage-2 overrides Stage-1.

Example

Stage-1:

```
SH = Non-Shareable
```

Stage-2:

```
S2SH = Inner Shareable
```

Final:

```
Inner Shareable
```

Hypervisor ensures SMP correctness.

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16.16 — Shareability and QNX/RTOS Behavior

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QNX uses:

- Inner Shareable for all normal memory
- Non-Shareable for device memory
- deterministic shareability rules
- strict SMP ordering

Partitioning

Each partition uses:

- Inner Shareable for shared memory
- Non-Shareable for I/O memory

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16.17 — Common Shareability Mistakes

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Mistake 1 — Mapping MMIO as Inner Shareable

Causes device malfunction.

Mistake 2 — Mapping RAM as Non-Shareable

Breaks SMP coherence.

Mistake 3 — Using Outer Shareable unnecessarily

Reduces performance.

Mistake 4 — Using Non-Shareable for DMA buffers

Breaks DMA.

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16.18 — Full Shareability Example

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Given descriptor:

```
SH = 11 → Inner Shareable
AttrIndx = 0 → Normal WB
```

Final behavior:

- cacheable
- coherent across cores
- atomic operations allowed
- prefetch allowed
- speculation allowed

Used for RAM.

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16.19 — Chapter Summary

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This chapter covered:

- shareability domains
- Stage-1 SH
- Stage-2 S2SH
- TCR shareability
- cache coherency
- atomic operations
- memory ordering
- DMA behavior
- TLB coherency
- walk cache coherency
- speculation
- prefetch
- Stage-1 vs Stage-2 merging
- QNX/RTOS behavior
- common mistakes
- full examples